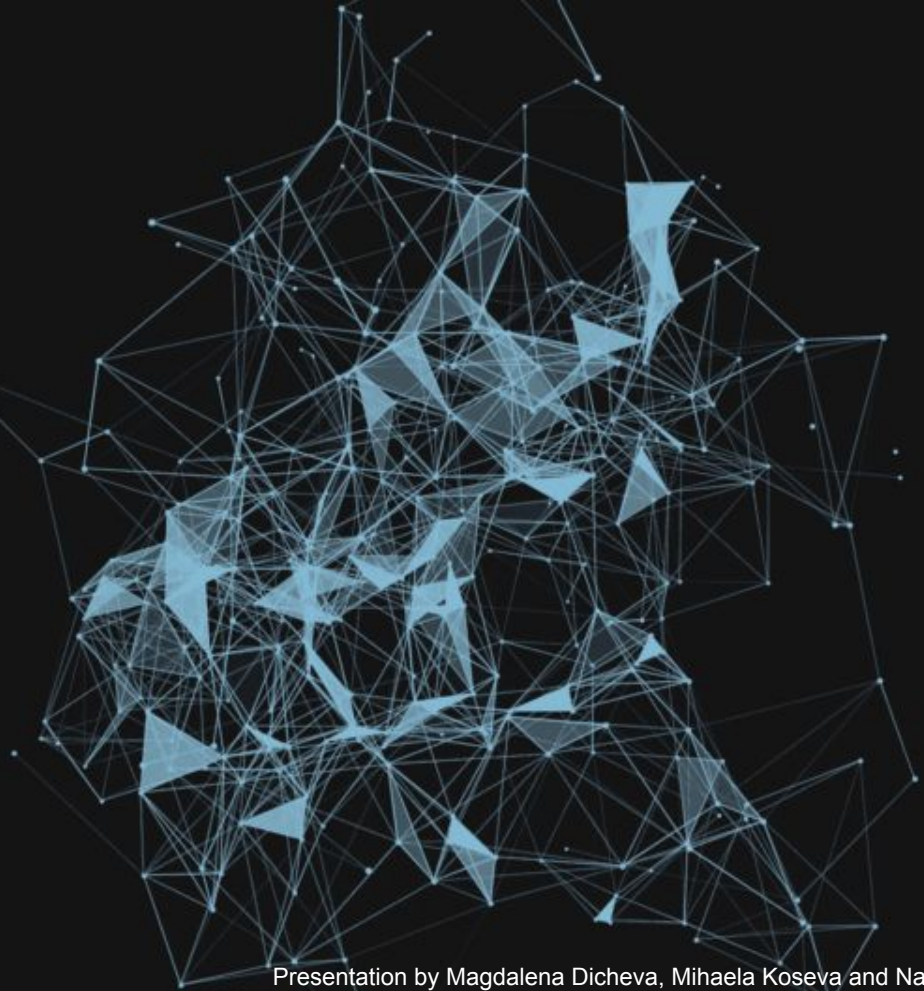


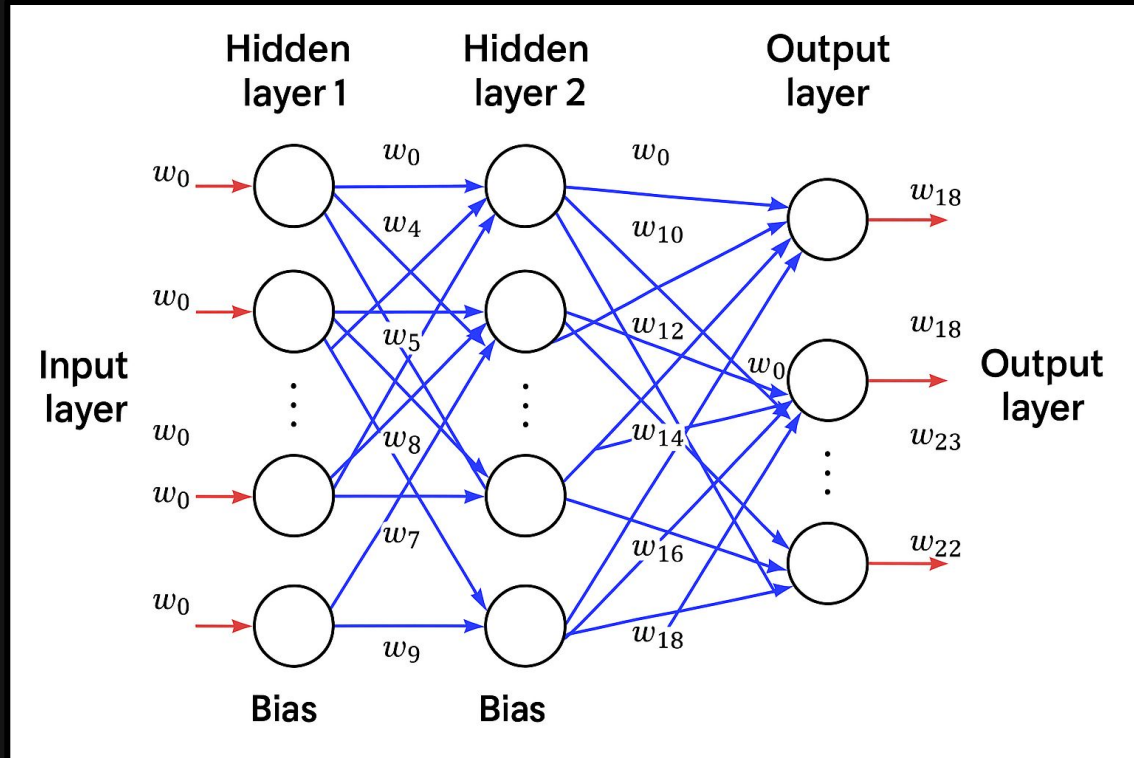
Невронни мрежи

SNN – Neuron impulses

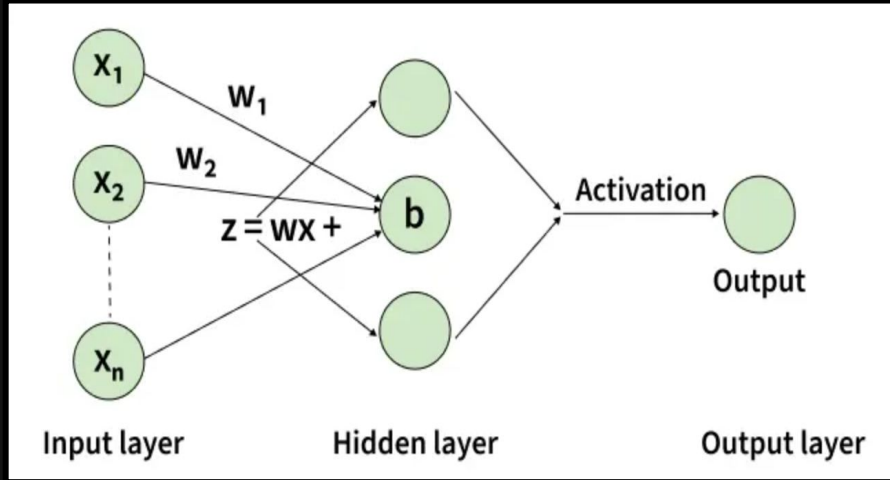


Presentation by Magdalena Dicheva, Mihaela Koseva and Nadya Radeva for the course 'Application of Mathematics In Modelling Real-life processes' At Faculty of Mathematics and Informatics, University 'St. Kliment Ohridski'

Невронни мрежи

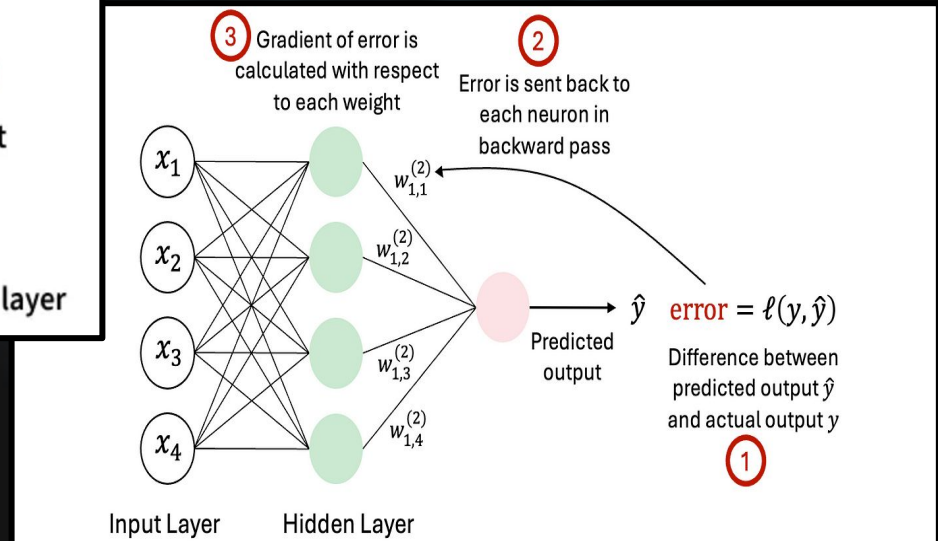


Forward propagation and Backpropagation

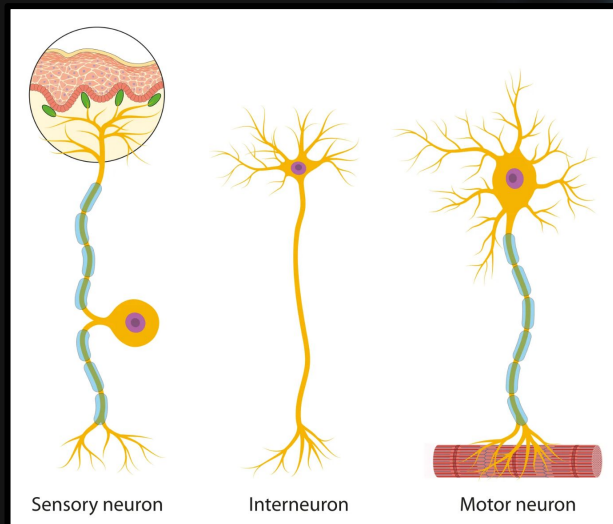
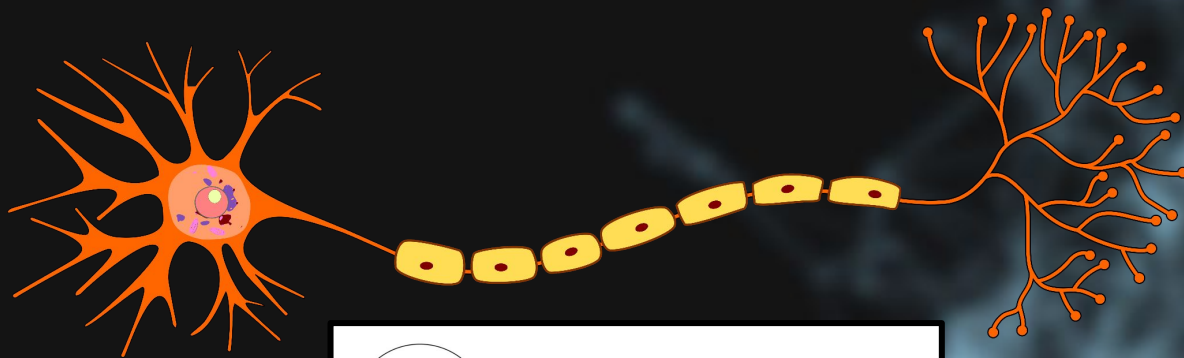


Forward propagation

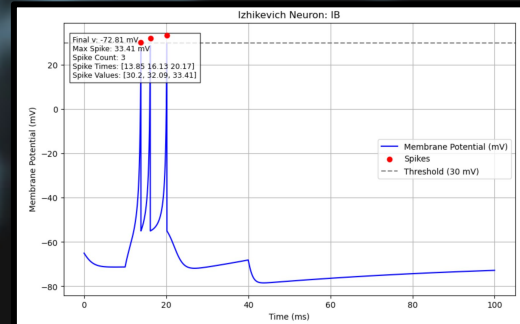
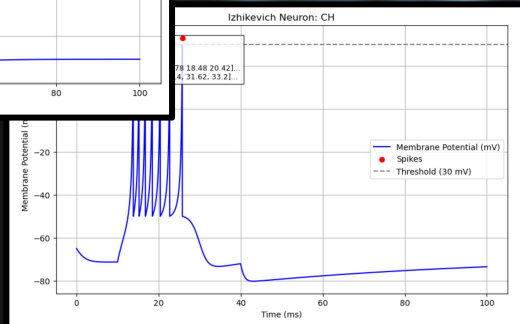
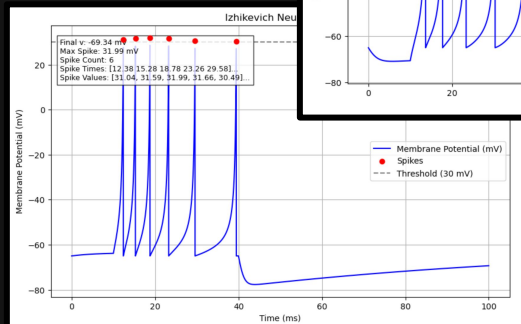
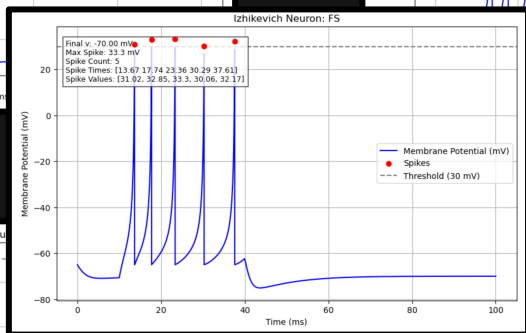
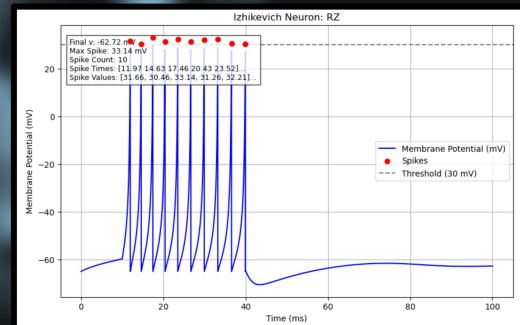
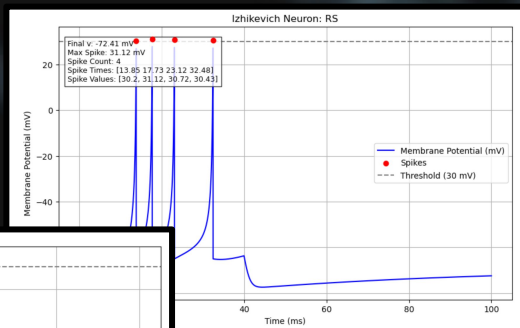
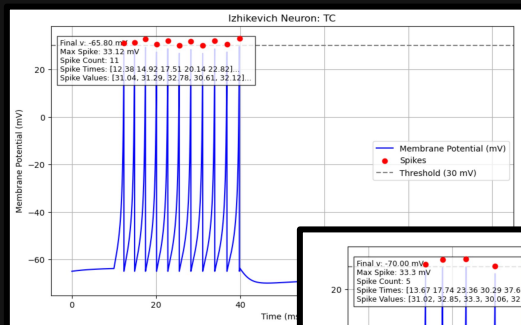
Backpropagation



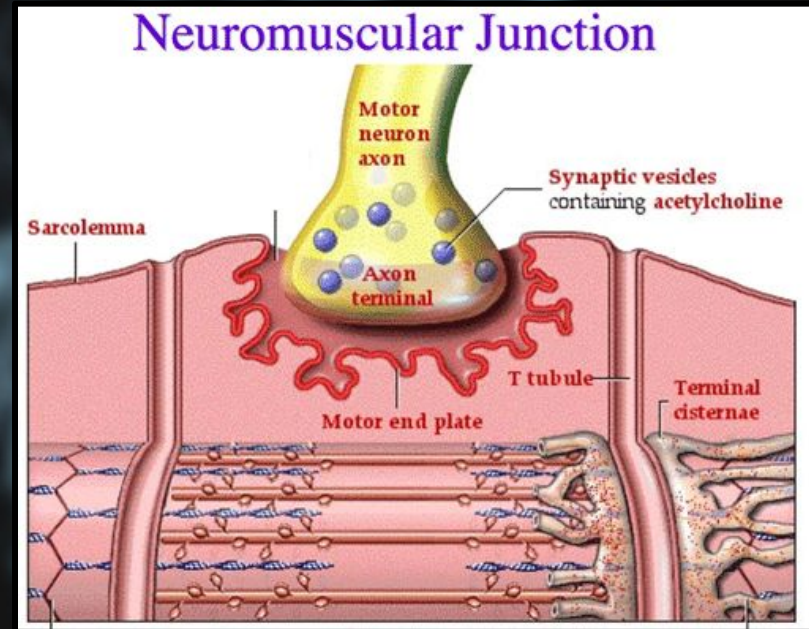
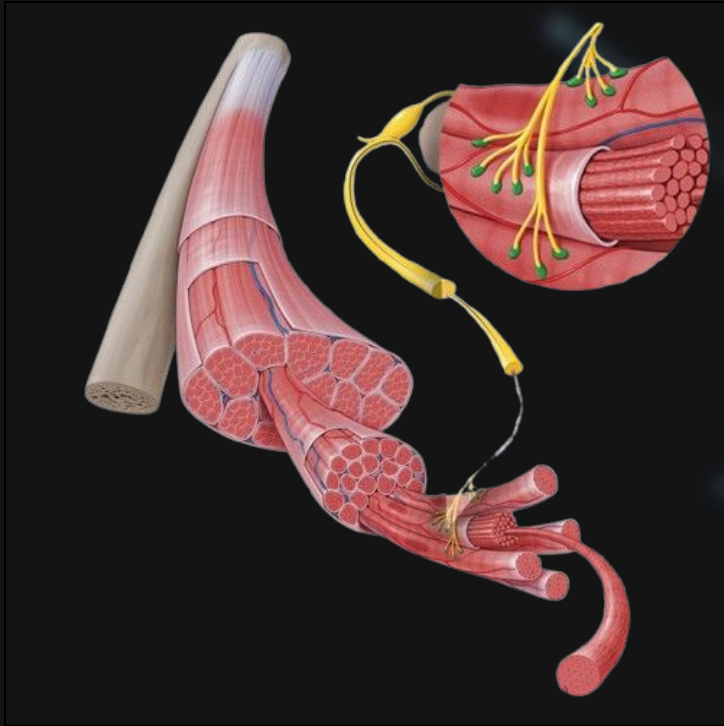
Биологични неврони



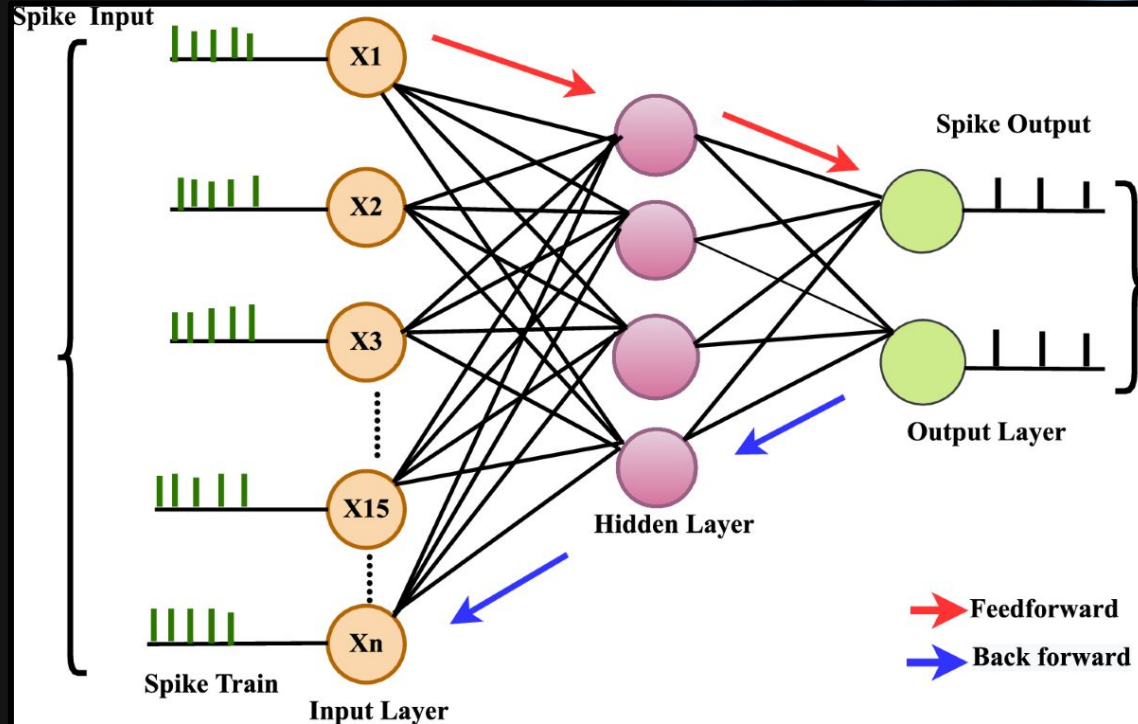
Видове неврони



Невромускулна активност



Spiking Neural Networks



Huxley-Hodgkin Model

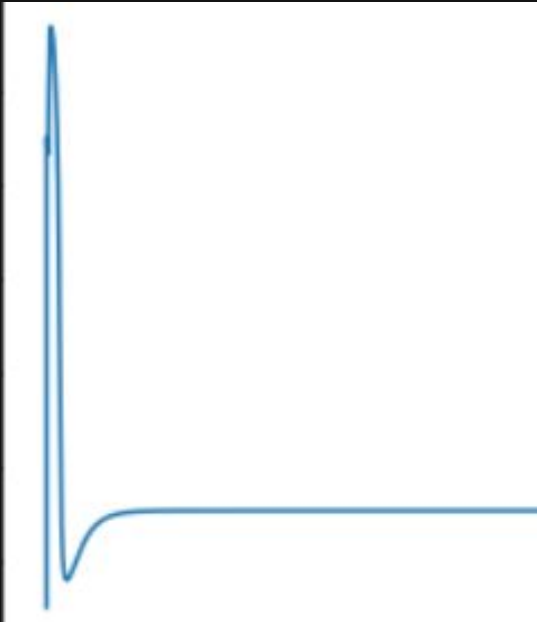
$$I = C_m \frac{dV_m}{dt} + \bar{g}_K n^4 (V_m - V_K) + \bar{g}_{Na} m^3 h (V_m - V_{Na}) + \bar{g}_l (V_m - V_l),$$

$$\frac{dn}{dt} = \alpha_n(V_m)(1 - n) - \beta_n(V_m)n$$

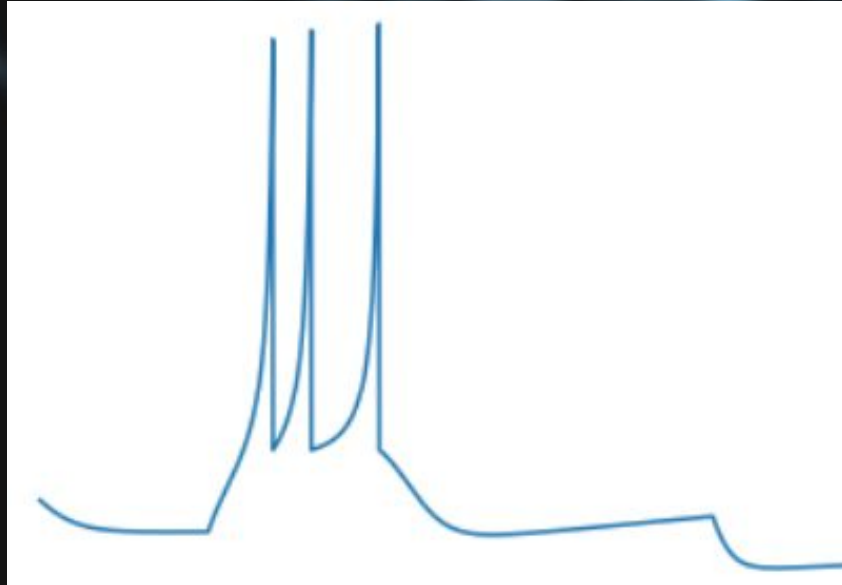
$$\frac{dm}{dt} = \alpha_m(V_m)(1 - m) - \beta_m(V_m)m$$

$$\frac{dh}{dt} = \alpha_h(V_m)(1 - h) - \beta_h(V_m)h$$

HH-model



Izhikevich model



Spiking Neural Networks - Coding + Decoding



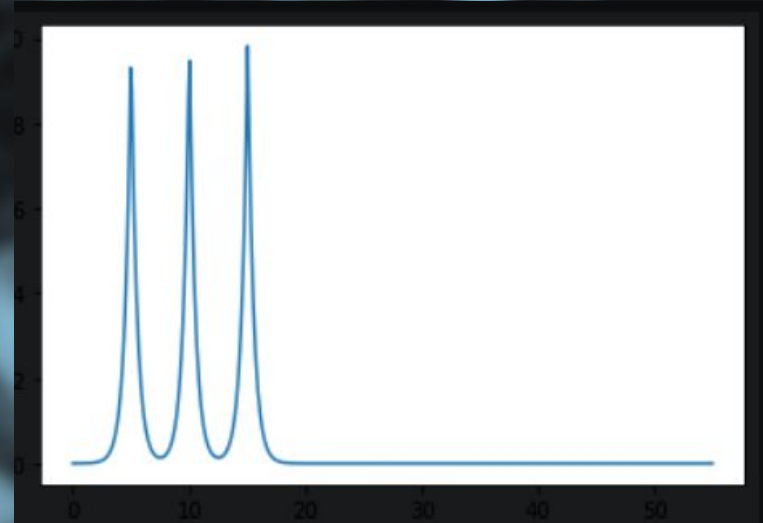
Algorithm

Normalization

$$x_{\text{norm}} = \frac{x - x_{\min}}{x_{\max} - x_{\min}}$$

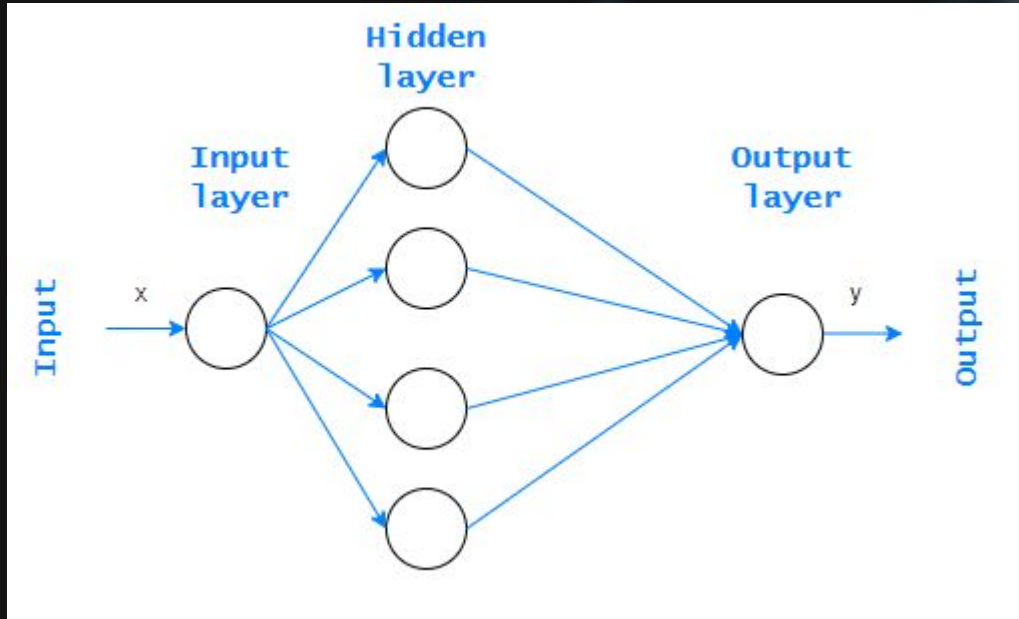
Frequency

$$\alpha = \alpha_{\min} + x (\alpha_{\max} - \alpha_{\min})$$



$$W e^{-\frac{|t-t_i|}{\tau}}$$

How SNN works?



Euler Method

$$y_{n+1} = y_n + h f(t_n, y_n)$$

Izhikevich model

$$\frac{du}{dt} = a(bv - u)$$

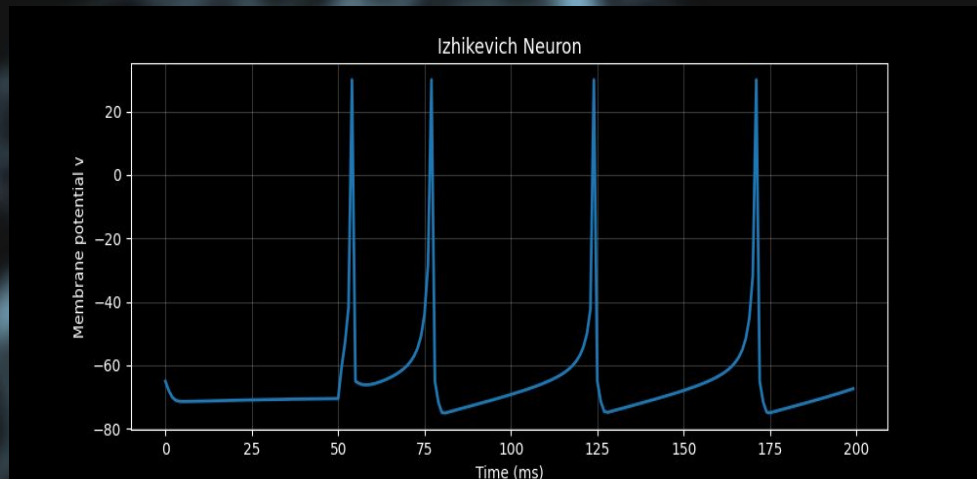
$$\frac{dv}{dt} = 0.04v^2 + 5v + 140 - u + I$$

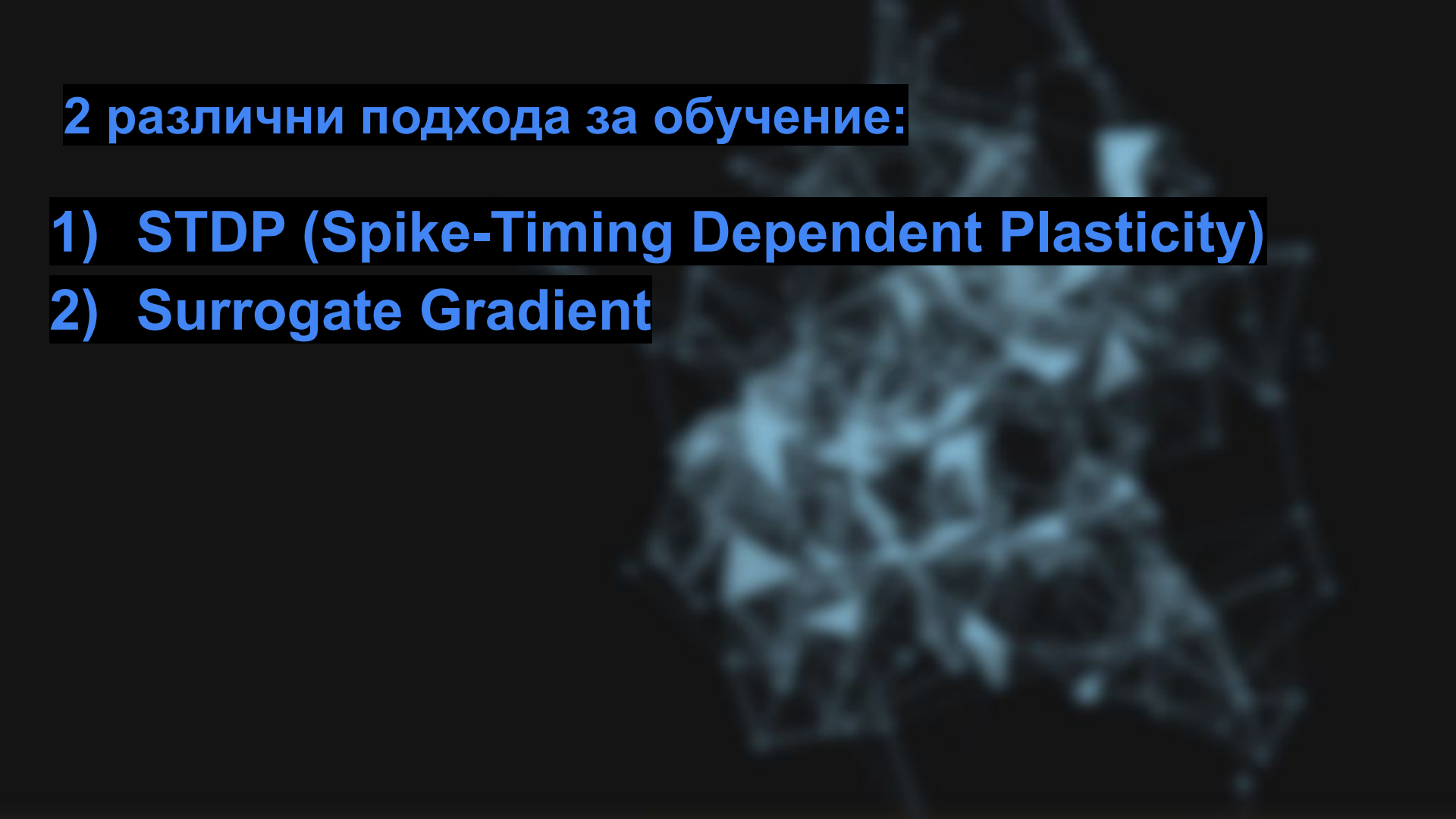
Моделът на *Izhikevich*

Параметър	Значение
a	скорост на възстановяване
b	чувствителност
c	reset на v
d	промяна на u след spike

$$\frac{dv}{dt} = 0.04v^2 + 5v + 140 - u + I$$

$$\frac{du}{dt} = a(bv - u)$$





2 различни подхода за обучение:

1) STDP (Spike-Timing Dependent Plasticity)

2) Surrogate Gradient

STDP (Spike-Timing Dependent Plasticity)

t = 0 ms

t = 1 ms

t = 2 ms

...

t = 100 ms

$v \geq 30$

...

$$I = w \cdot spike$$

$$\Delta t = t_B - t_A$$

$$\Delta t = 5$$

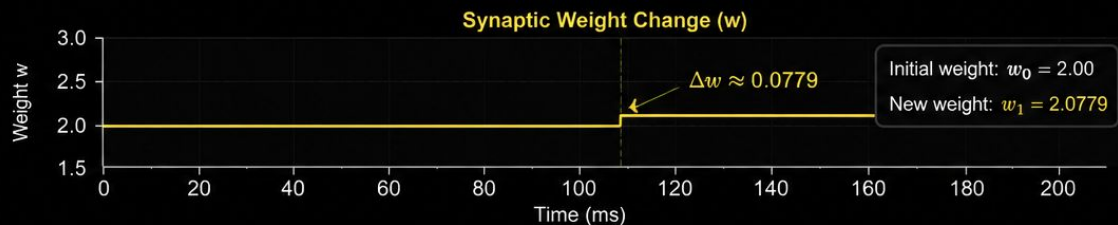
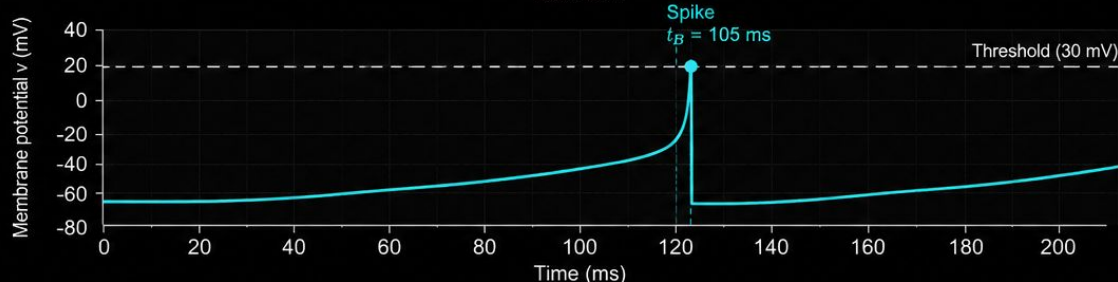
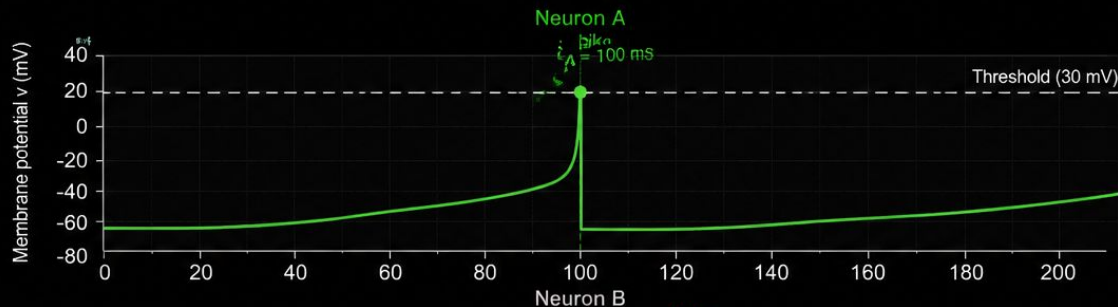
$$\Delta w = 0.1e^{-\Delta t/20}$$

$$w = w + \Delta w$$

STDP Simulation Example

A $\rightarrow$ B

A fires at $t_A = 100$ ms, B fires at $t_B = 105$ ms



STDP Rule

$$\Delta t = t_B - t_A$$

$$\Delta w = \begin{cases} A_+ e^{-\Delta t / \tau_+}, & \Delta t > 0 \\ -A_- e^{\Delta t / \tau_-}, & \Delta t < 0 \end{cases}$$

Calculation

$$t_A = 100 \text{ ms}$$

$$t_B = 105 \text{ ms}$$

$$\Delta t = 105 - 100 = 5 \text{ ms}$$

$$\Delta w = A_+ e^{-\Delta t / \tau_+}$$

$$A_+ = 0.1, \tau_+ = 20 \text{ ms}$$

$$\Delta w = 0.1 \cdot e^{-5/20} = 0.1 \cdot e^{-0.25}$$

$$\Delta w \approx 0.0779$$

Summary

- Neuron A fires at 100 ms
- Neuron B fires at 105 ms
- $\Delta t = 5 \text{ ms} > 0 \rightarrow$ A before B
- The synaptic weight is **potentiated**
- w increases from 2.00 to **2.0779**

If A fires before B ($\Delta t > 0$), the connection A $\rightarrow$ B is strengthened.

If B fires before A ($\Delta t < 0$), the connection is **weakened**.

Surrogate Gradient

$$I = w \cdot \text{spike}$$

$$S(v) = \begin{cases} 1, & v \geq 30 \\ 0, & v < 30 \end{cases} \quad \begin{array}{l} \text{target} = 1 \\ \text{output} = 0/1 \end{array}$$

$$L = (1 - 0)^2$$
$$L = 1$$



$$S(v) \approx \sigma(v)$$



$$L = [1 - \sigma(v-30)]^2$$



$$w_{\text{new}} = w - \eta \frac{\partial L}{\partial w}$$

Сравнение на Izhikevich и Hodgkin–Huxley модел

1. Математика
2. Скорост
3. Биологична точност
4. Какво описват?
5. Сложност
6. Къде се използват?

